

Solve the problem below. After you are done, rename the file containing your source code as your *StudentId.c* (For example, if your student ID is *2005001*, the name of your file should be *2005001.c*). Then submit that file to Moodle. Make sure you submit a file containing your source code.

***Failure to follow these instructions will result in penalties.**

Problem Description

1. Write a C program that has a **recursive function** named "**remove_n_digits**" that removes the **last "c"** occurrences of a digit "**d**" from an integer "**n**". [10]

Function Prototype & Sample Call

```
int remove_n_digits(int n, int d, int c)
{
    // TODO
}

// result = 7901510
int result = remove_n_digits(79015910, 9, 1);
```

Sample Calls to Function	Sample Return Value
remove_n_digits(79015 9 10, 9 , 1)	7901510
remove_n_digits(7 9 015 9 10, 9 , 2)	701510
remove_n_digits(777 , 7 , 4)	0
remove_n_digits(777, 5 , 4)	777
remove_n_digits(123 4 5, 4 , 3)	1235

N.B.:

- ★ You **must** use recursion to implement this function. In particular, your function **can not** contain the keywords **for** or **while** in its body.
- ★ Your **main** function **can not** have any other processing statements except for taking input, calling the function and displaying output.
- ★ You **can not** use any library function for this task (other than **I/O**).
- ★ You **can not** use any **array** or **pointer** while solving this problem.
- ★ You **can** assume that the provided input will **always** be valid.